AI Tool Prompt Engineering Report

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AI Tool Overview:

Perplexity AI is a tool mainly used for information retrieval and search, helping users find reliable answers by gathering data from different sources. I chose this tool because I have already used it before when looking for references, and I found its results to be mostly accurate and helpful. What makes it useful is that it does not rely on just one source, but instead combines information from many, which makes the answers more complete and trustworthy. Because of this, it is a good tool for students who need quick and reliable information for their studies.

Original Prompt: Tell me everything that there is needed to know about mechanical engineering, specifically in the manufacturing industry.

Result of Original Prompt:

START

Mechanical engineering in manufacturing is about designing, operating, and improving the systems, machines, and processes that turn raw materials into reliable, cost-effective products at scale. Below is a structured overview of what you "need to know" for this domain.

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1. What mechanical engineers actually do in manufacturing

Mechanical engineers in manufacturing combine design, analysis, and shop-floor work to keep production efficient, safe, and high quality.

Key responsibilities:

- Design or adapt products for manufacturability (DFA/DFM, tolerances, materials).
- Develop and improve manufacturing processes and workflows (line layouts, cycle time, bottleneck removal).
- Specify, design, or select production equipment (machine tools, fixtures, conveyors, robots).

- Oversee and troubleshoot production (quality issues, downtime, machine failures).
- Implement Lean, Six Sigma, and continuous improvement projects to reduce waste and defects.
- Coordinate with design, quality, maintenance, purchasing, and suppliers.
- Prepare technical documentation: process sheets, work instructions, tooling drawings, maintenance procedures.

Typical job titles:

- Mechanical engineer (plant or production)
- Manufacturing engineer / process engineer
- Tooling / methods / industrialization engineer
- Maintenance / reliability engineer (mechanical focus)

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2. Core technical foundations

To work in manufacturing, a mechanical engineer must apply several core knowledge areas.

Key fundamentals:

- Mechanics and machine design: stress/strain, fatigue, shafts, gears, bearings, fasteners, mechanisms for equipment and tooling.
- Materials science: metals, polymers, composites, and their properties, heat treatment, and how processing affects performance.
- Thermodynamics and heat transfer: furnaces, ovens, cooling, power and compressed-air systems, HVAC for facilities.
- Fluid mechanics: pumps, piping, hydraulics, pneumatics used in machines and plants.
- Controls and mechatronics basics: sensors, actuators, PLCs, motion control, robotics integration with mechanical systems.
- Manufacturing science: cutting mechanics, forming, casting solidification, joining metallurgy, surface engineering.

Supporting skills:

- Engineering drawing/GD&T and CAD (3D modeling, detail drawings).
- Statistics and quality tools (SPC, capability, DOE).
- Basic economics and cost analysis (capital vs operating cost, ROI, cost per part).

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3. Major manufacturing processes you must know

Mechanical engineers in manufacturing need at least a working understanding of the main process families and when to use each.

3.1 Process families

- Casting: melt metal and pour into a mold (sand, die casting, investment casting).
- Forming/metalworking: deform solid material (forging, rolling, extrusion, drawing, stamping).
- Machining (material removal): turning, milling, drilling, grinding, EDM for precision features.
- Joining: welding, brazing, soldering, adhesive bonding, mechanical fasteners (bolts, rivets, screws, press-fit).
- Heat treatment: controlled heating and cooling to change hardness, strength, toughness.
- Surface processes: cleaning, coating, plating, painting, anodizing, thin-film deposition.
- Additive manufacturing: 3D printing metals and plastics for prototypes, tooling, and sometimes production.

3.2 Example: joining processes

Joining is critical in assembly and often drives cost and reliability.

- Welding: permanent metallurgical joint, suitable for structural and pressure parts.
- Brazing/soldering: lower-temperature joining with filler metal, common for heat exchangers, electronics.
- Adhesive bonding: distributes stress, useful for composites and dissimilar materials.
- Mechanical fasteners: bolts, screws, rivets for serviceability and temporary joints.

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4. Manufacturing systems, quality, and productivity

Beyond single processes, mechanical engineers design and improve systems: the full production flow.

4.1 Production systems and layout

- Line vs batch vs job-shop vs cellular layouts, depending on product mix and volume.
- Material handling: conveyors, automated guided vehicles, cranes, pallets, ergonomics.
- Automation: CNC machines, robots, flexible manufacturing systems, integrated with sensors and controls.

4.2 Lean, Six Sigma, and continuous improvement

Mechanical engineers apply structured methods to cut waste and variation.

- Lean: eliminate non-value-adding activities (transport, waiting, overproduction, defects, motion, inventory, over-processing).
- Tools: value stream mapping, 5S, SMED (quick changeover), standardized work, Kanban.
- Six Sigma: statistical problem-solving to reduce defects (DMAIC, capability indices, DOE).

4.3 Quality and metrology

- Tolerancing and fit selection to balance performance and manufacturability.
- Inspection methods: gauges, CMMs, optical systems, non-destructive testing.
- Statistical process control (SPC) and process capability (Cp/Cpk) to keep production stable.

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5. Roles, skills, and career focus in manufacturing

Mechanical engineers in manufacturing can specialize along several tracks.

5.1 Typical specialized roles

Focus area	What you mainly do	Key skills needed
Manufacturing / process	Design and optimize processes, line balancing, capacity planning, tooling layouts.	Process science, simulation, Lean, data analysis.
Production / plant engineer	Support day-to-day production, troubleshoot equipment, coordinate maintenance.	Hands-on mechanical skills, problem-solving, communication.
Tooling / fixtures	Design jigs, fixtures, cutting tools, dies, assembly aids.	Machine design, GD&T, machining/forming know-how.
Quality / reliability	Investigate defects, root cause analysis, FMEA, reliability of components and systems.	Statistics, failure analysis, standards knowledge.
Automation / mechatronics	Integrate robots, sensors, and control systems into lines.	Mechatronics, controls, programming basics.
Maintenance / reliability	Plan preventive maintenance, analyze failures, extend asset life.	Rotating equipment, lubrication, vibration basics.

5.2 Soft skills and tools

- Communication across departments and with operators is essential for practical solutions.

- Familiar tools: CAD, CAM, simulation (FEA/CFD basics), data/statistical software, PLC/robot interfaces.
- Documentation and project management to deliver changes safely and on schedule.

If you tell me your current level (student vs working engineer) and which part of manufacturing you're most interested in (design, process, automation, quality, etc.), I can turn this into a focused learning roadmap with concrete next steps.

END

AI Assisted Prompt (ChatGPT): "Provide a comprehensive, structured overview of mechanical engineering in the manufacturing industry. Cover core concepts (materials, thermodynamics, mechanics, manufacturing processes), key technologies (CNC, automation, robotics), industry workflows, common tools and software, and real-world applications. Include practical examples and explain how these concepts are used in actual manufacturing settings."

Result of AI Assisted Prompt:

START

Mechanical engineering drives manufacturing by applying physics, materials science, and design principles to create efficient production systems and products. It optimizes processes from raw materials to finished goods, enhancing quality, reducing waste, and integrating advanced tech.

Core Concepts

Materials science guides selection of metals, polymers, ceramics, and composites for strength, durability, and cost in manufacturing. Thermodynamics and heat transfer manage energy in processes like forging or cooling, ensuring efficient thermal systems. Mechanics covers statics, dynamics, and structural analysis for designing load-bearing parts, while manufacturing processes include machining, casting, welding, and forming to shape components precisely.

Key Technologies

CNC machines use computer control for precise cutting, milling, and turning, enabling complex parts with tight tolerances. Automation streamlines workflows via robotic arms for loading/unloading and pallet systems, reducing downtime. Robotics, including SCARA and collaborative types, handles assembly, welding, and inspection, often integrated with AI vision for adaptability.

Industry Workflows

Workflows start with planning and scheduling to align resources and timelines. Execution involves material processing, machining, assembly, and quality checks using statistical controls. Completion includes routing, approvals, and output delivery, with feedback loops for continuous improvement like lean principles.

Tools and Software

CAD tools like SolidWorks, CATIA, and Fusion 360 enable 3D modeling and simulation. CAM software such as Mastercam generates CNC toolpaths for manufacturing. FEA/CFD tools like ANSYS analyze stress, fluids, and thermodynamics before production.

Real-World Applications

In automotive manufacturing, Toyota uses robotics and lean principles for efficient assembly lines, cutting costs and ensuring consistency. Aerospace applies advanced materials and CNC for lightweight, durable aircraft parts via simulation-optimized designs. Medical robotics leverages precision mechanics for surgical tools, meeting hygiene and reliability standards through automated processes.

END

Comparison and Analysis:

In comparing the outputs from the original and improved prompts, I noticed several key differences in terms of detail, organization, and overall usefulness. The original prompt produced a response that was more detailed and informative. It presented ideas in a clearer and more structured way, making it easier to follow and

understand. On the other hand, the improved prompt resulted in a more compact output. While it was organized like an essay and easier to read in one go, it did not go as deep into the topic. It felt more like a summary rather than a full discussion.

When it comes to usefulness and accuracy, I believe the original prompt was more helpful, especially for someone who already has some background knowledge about the topic. The level of detail made it easier to learn more and understand the subject better. It was also more engaging to read because it explained concepts more thoroughly. In contrast, the improved prompt seemed more suitable for beginners. It provided a clear starting point, but it lacked the depth needed for deeper learning.

The improved prompt did have some advantages, particularly in terms of clarity. The response was easier to understand, especially for someone who is new to the topic. However, this clarity came at the cost of depth. The information presented was more surface-level, and it did not explore the topic in as much detail as the original output. Despite this, the relevance of the response was still maintained, as it stayed focused on the main objective and did not include unnecessary information.

Overall, the differences in the outputs were mainly due to how the prompts were structured. The improved prompt aimed to simplify and organize the response, which made it better for beginners, while the original prompt allowed for a more detailed and in-depth explanation. From this, I learned that prompt engineering is not as simple as it seems. Creating an effective prompt requires clear intention and a good understanding of the desired outcome. The way a prompt is written can greatly affect the quality, clarity, and depth of the response.

Appendix:

Can you improve this prompt "Tell me everything that there is needed to know about mechanical engineering, specifically in the manufacturing industry."

Your original prompt is a bit too broad, so the answer you get will likely be unfocused or shallow. A stronger prompt should guide depth, structure, and level.

Here are improved versions depending on what you want:

- **Best Overall (Structured + Deep)**

"Provide a comprehensive, structured overview of mechanical engineering in the manufacturing industry. Cover core concepts (materials, thermodynamics, mechanics, manufacturing processes), key technologies (CNC, automation, robotics), industry workflows, common tools and software, and real-world applications. Include practical examples and explain how these concepts are used in actual manufacturing settings."

- **If you want it tailored for learning (student-friendly)**

"Explain mechanical engineering in the manufacturing industry in a way suitable for a 2nd-year engineering student. Break down the key subjects I need to master, the important manufacturing processes, and how they connect in real-world applications. Include examples, common mistakes, and tips for understanding the field deeply."

- **If you want career-focused insight**

"Give a detailed guide to mechanical engineering in the manufacturing industry, including required skills, tools, typical job roles, industry trends, and how a student can prepare to work in this field. Include practical advice and real-world examples."

- **If you want expert-level depth**

"Provide an in-depth technical overview of mechanical engineering in the manufacturing industry, including advanced manufacturing processes, material selection, process optimization, automation systems, and quality control methods. Discuss how these elements integrate in modern production systems."

If you want, I can combine all of these into one "perfect prompt" tailored exactly to your level and goals (since you're a mech eng student).

From the Original Prompt:

10 sources

Follow-ups

- Create a comprehensive PDF guide on mechanical engineering in manufactu... Computer
- What are the latest advancements in automation for manufacturing mechanical engineers
- How do mechanical engineers apply CAD/CAM in manufacturing
- What skills are essential for mechanical engineers in manufacturing careers
- Compare casting vs machining processes in manufacturing

From the AI-Assisted Prompt:

18 sources

Follow-ups

- Create a professional PowerPoint slide deck overviewing mechanical engine... Computer
- How does AI integration enhance mechanical engineering in manufacturing
- Compare CAD software like Solidworks CATIA and PTC Creo for manufacturing
- What are common career paths for mechanical engineers in manufacturing
- Challenges and solutions in implementing CNC robotics in factories